Om Rajendra Kathalkar

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PROFESSIONAL SUMMARY

PhD researcher in Computer Science at the National Technical University of Athens, focusing on efficient and deployable AI systems, particularly inference optimization under real-world constraints. My work spans computer vision, edge AI, and network-aware systems through academic research and industry collaborations. Alongside research, I am deeply interested in teaching and mentoring, and enjoy helping learners build strong foundations in AI through clear, practical guidance.

EDUCATION

PhD in Computer Science Fall 2025 – Present
National Technical University of Athens (NTUA), Greece
Advisor: Dr. Konstantinos Tserpes

M.S. by Research, Electronics and Communication Engineering 2021 – 2025
International Institute of Information Technology (IIIT), Hyderabad
Advisor: Dr. Sachin Chaudhari
Thesis: Camera-Based Deep Learning Framework for AQI Estimation: Dataset and Methodology

B.E. in Electronics and Telecommunication Engineering
St. Vincent Pallotti College of Engineering and Technology, Nagpur

RESEARCH EXPERIENCE

Research Associate Jan 2026 – Present
Institute of Communication and Computer Systems (ICCS), NTUA, Greece

- Research on **efficient and autonomous execution of AI models**, focused on edge inference and scalable AI systems, in collaboration with NTUA research groups.
- Contributing to **Horizon Europe** initiatives on AI-driven optimization, real-world deployment constraints, and reproducible research pipelines.
- Collaborating with interdisciplinary academic and industry teams to translate theoretical models into **deployable AI solutions** for connected systems.

Applied AI Researcher Dec 2024 – Dec 2025
Ericsson Research, Bangalore

- Contributed to the European project PANDORA on network-aware path planning for autonomous mobile robots (AMRs).
- Developed algorithms combining real-time decision-making and semantic understanding for reliable navigation in dynamic industrial environments under network constraints.

Research Scholar Aug 2021 – Oct 2025
Signal Processing and Communications Research Centre, IIIT Hyderabad

- Research with Dr. Sachin Chaudhari on edge AI for IoT and computer vision towards a smarter Internet of Things.
- Led development of a sensor-free AQI estimation framework using camera data, reducing dependence on costly, failure-prone sensor networks; created **TRAQID** (26,678+ traffic-related images with co-located weather data), a Spotlight paper at ICVGIP 2024 and among the conference's most downloaded works.
- Proposed **AQIFormer**, a transformer-based multi-view model reaching ~90% AQI classification accuracy, including 88% at night and 81.67% in cross-city transfer.

TEACHING EXPERIENCE

Gen AI–ML Instructor Apr 2024 – Present

TalentSprint — Part of Accenture

- Teach machine learning, deep learning, and NLP for programs from IIIT Hyderabad, IISc Bangalore, and the University of Michigan.
- Mentored **250+ industry professionals** on **Agentic and Generative AI**, including learners with 15+ years of experience, through hands-on workflows and applied AI systems.

AI Instructor Apr 2025 – Jan 2026

iHub-Data, Hyderabad

- Delivered modules on classification, regression, clustering, PCA, neural networks, and ML/DL applications in a 24-week AI/ML student training program.
- Ran hands-on tutorials, practical evaluations, and performance tracking for 50+ engineering students.

Instructor, Embedded Systems Workshop Monsoon 2024

Instructor, ML for IoT Workshop Monsoon 2023

Instructor, Machine Learning KEIT, Spring 2022

PUBLICATIONS

JOURNAL PAPERS

[J1] **Om Kathalkar**, H. Hajj Hassan, A. Kattepur, G. Bouloukakis. "Network-aware path planning for AMRs: a generalizable framework with spatial-aware adaptation." *Autonomous Robots* (Springer Nature), under review.

CONFERENCE PAPERS

[C5] **Om Kathalkar**, N. Nilesh, S. Chaudhari, A. Namboodiri. "AQIFormer: a transformer-based multi-view architecture for cross-city air quality classification." *ICVGIP*, 2024.

[C4] **Om Kathalkar**, H. Hajj Hassan, A. Kattepur, G. Bouloukakis. "Network-aware path planning for autonomous mobile robots in industrial environments." *IEEE ANTS*, Delhi, 2025. (Acceptance rate 32.8%)

[C3] **Om Kathalkar**, N. Nilesh, S. Chaudhari, A. Namboodiri. "TRAQID — traffic-related air quality image dataset." *ICVGIP*, Bangalore, 2024. **(Spotlight)** [code]

[C2] S. Sara, A. Rebeiro-Hargrave, **Om Kathalkar**, S. Chaudhari. "Protocol for hunting PM2.5 emission hot spots in cities." *1st Int. Workshop on Advances in Environmental Sensing Systems for Smart Cities (EnvSys)*, Helsinki, 2023. [code]

[C1] K. S. Viswanadh, **Om Kathalkar**, S. Chaudhari. "CV and IoT-based remote triggered labs: use case of conservation of mechanical energy." *FiCloud*, Rome, 2022. [code]

PATENTS

[P3] Indian patent filed: "System for determining Air Quality Index (AQI) in urban environments." App. No. 202541073518, May 2025.

[P2] Indian patent published: "System and method for determining air quality by processing environmental and traffic-related visual data." Pub. No. 202441043706, May 2024.

[P1] US patent published: "System and method for implementing an experiment remotely and determining an output using a computer vision model." Pub. No. US20240078641A1, March 2023.

REVIEWING SERVICES

Reviewer — Conference on Neural Information Processing Systems (NeurIPS) 2026

Reviewer — Asian Conference on Computer Vision (ACCV)

Reviewer — IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)

Reviewer — International Conference on Future Internet of Things and Cloud (FiCloud)

Reviewer — IEEE INDICON

RESEARCH PROJECTS

TURING *(Horizon Europe)* 2026 – Present

Efficient and autonomous execution of AI algorithms, emphasizing **scalable inference, trustworthy AI, and deployable pipelines** for industrial systems.

PANDORA *(Horizon Europe)* 2024 – 2025

Network-aware and trustworthy AI for autonomous mobile robots: robust path planning, communication-aware decision-making, and reliable operation in industrial environments.

IoT-Enabled Smart Cities: Pollution, Health and Governance 2019 – 2023

Funded by PRIF and DST, India · IIIT Hyderabad. Designed **image-based air quality monitoring** integrating traffic visuals and environmental cues for scalable, low-cost AQI estimation.

Remote Labs Platform 2021 – 2022

Funded by PRIF and DST, India · IIIT Hyderabad. Built and supported a large-scale remote laboratory platform to **democratize STEM education** for thousands of rural and underrepresented students across India.

SKILLS

Areas of interest	Machine & deep learning, edge AI, Internet of Things, NLP, agentic & generative AI
Tools	PyTorch, TensorFlow, NVIDIA Sionna, CUDA, Django, scikit-learn, LaTeX, n8n
Hardware	Jetson Xavier NX, Raspberry Pi, Arduino, ESP-series microcontrollers

AWARDS AND HONORS

- **iHub-Data Research Fellowship** (Aug 2023) for MS by Research at IIIT Hyderabad — full research funding and tuition, awarded on academic merit and research potential.
- **Winner**, International Environmental Sensing Project Competition (Mar 2023), University of Helsinki — top position among teams from multiple countries.
- **AIR-1 Gold Award**, e-Yantra Ideas Competition, IIT Bombay — national competition with 100+ engineering college teams.
- **Student of the Year** (May 2023) among a batch of 1000+ undergraduates at SVPCET, Nagpur.
- **Best Project Award** (Dec 2022) for the undergraduate final-year project.
- **Student of the Department (ETC)**, SVPCET, Nagpur, for research contributions and academic excellence.

LETTERS OF RECOMMENDATION

Dr. Sachin Chaudhari, Associate Professor, IIIT Hyderabad, India. [letter]

Dr. Ajay Kattepur, Senior Researcher, Ericsson Research, Bangalore, India. [letter]

Dr. Georgios Bouloukakis, Assistant Professor, University of Patras, Greece. [letter]

Dr. Houssam Hajj Hassan, Senior Researcher, Orange Research, France. [letter]